

CMI Entrance Examination

Mock Test - Part B (Solutions)

Solution to Question 1

- (a) Let the tiles be denoted by V (vertical 2×1 , length 1), H (two horizontal 1×2 stacked, length 2), and S (2×2 square, length 2). The restriction is that S cannot be immediately followed by another S . Let C_n be the total valid tilings of length n .

Base Cases:

- $n = 1$: Only V fits. $\implies C_1 = 1$.
- $n = 2$: Combinations are VV, H, S . $\implies C_2 = 3$.
- $n = 3$: Valid sequences are VVV, VH, HV, VS, SV . $\implies C_3 = 5$.
- $n = 4$: To tile a 2×4 board, we build upon smaller boards. We will systematically count them using recurrence states below, which yields $C_4 = 10$.

Let S_n be tilings ending in S , and N_n be tilings ending in V or H . We have $C_n = S_n + N_n$. To form N_n , we can append V to any valid tiling of length $n - 1$, or H to any valid tiling of length $n - 2$. Thus, $N_n = C_{n-1} + C_{n-2}$. To form S_n , we must append S to a tiling of length $n - 2$ that does *not* end in S . Thus, $S_n = N_{n-2}$. Substituting S_n : $C_n = N_{n-2} + N_n = (C_{n-3} + C_{n-4}) + (C_{n-1} + C_{n-2})$. Therefore, the recurrence relation is:

$$\boxed{C_n = C_{n-1} + C_{n-2} + C_{n-3} + C_{n-4} \quad \text{for } n \geq 5}$$

Using $C_0 = 1, C_1 = 1, C_2 = 3, C_3 = 5$, we get $C_4 = 5 + 3 + 1 + 1 = 10$.

- (b) To find a closed-form expression, we use generating functions. Let $C(x) = \sum_{n=0}^{\infty} C_n x^n$. Using the sub-sequences $N(x)$ and $S(x)$: $S(x) = x^2 N(x)$ and $N(x) - 1 = xC(x) + x^2 C(x) = (x + x^2)C(x)$. Since $C(x) = S(x) + N(x) = (1 + x^2)N(x)$, we substitute $N(x)$: $\frac{C(x)}{1+x^2} - 1 = (x + x^2)C(x) \implies C(x)[1 - (x + x^2)(1 + x^2)] = 1 + x^2$.

$$\boxed{C(x) = \frac{1 + x^2}{1 - x - x^2 - x^3 - x^4}}$$

The closed form of C_n is the coefficient of x^n in the Taylor expansion of this rational function, which can be expressed explicitly as a linear combination of the powers of the roots of the characteristic equation $\alpha^4 - \alpha^3 - \alpha^2 - \alpha - 1 = 0$.

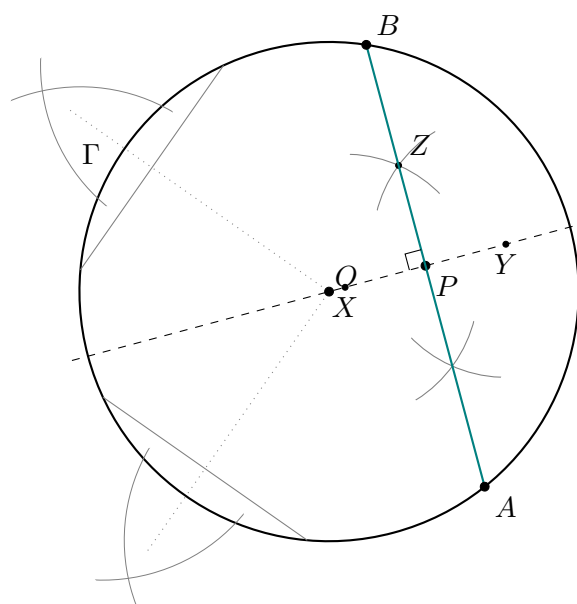
- (c) Let y trace the number of S tiles. The generating function becomes: $S(x, y) = yx^2 N(x, y)$ and $N(x, y) - 1 = (x + x^2)C(x, y)$. $C(x, y) = (1 + yx^2)N(x, y) \implies C(x, y) = \frac{1 + yx^2}{1 - x - x^2 - yx^3 - yx^4}$. To restrict to an even number of squares, we take the even part with respect to y : $D(x) = \frac{1}{2} (C(x, 1) + C(x, -1)) = \frac{1}{2} \left(\frac{1 + x^2}{1 - x - x^2 - x^3 - x^4} + \frac{1 - x^2}{1 - x - x^2 + x^3 + x^4} \right)$. Cross-multiplying and simplifying the numerators and denominators:

$$\boxed{D(x) = \frac{1 - x - x^2 + x^5 + x^6}{1 - 2x - x^2 + 2x^3 + x^4 - x^6 - 2x^7 - x^8}}$$

This generating function completely defines the recurrence for D_n .

Solution to Question 2:(a) **Construction Steps:**

1. Draw any two non-parallel chords on the circle Γ .
2. Using the compass, construct the perpendicular bisectors of both chords. Mark their intersection point as O . This is the center of Γ .
3. Draw the straight line OP passing through the center O and the given point P .
4. Using the compass, place the needle on P and mark two points X and Y on the line OP such that $XP = PY$.
5. Open the compass to a radius greater than XP . Draw intersecting arcs from centers X and Y to find point Z .
6. Draw the line ZP . This line is perpendicular to OP at P .
7. Let the line ZP intersect the circle Γ at points A and B . The chord AB is the required chord.



Proof: In a circle, a line passing through the center is perpendicular to a chord if and only if it bisects the chord. By our construction in step 6, the chord AB is perpendicular to the radius OP at P . Therefore, the perpendicular from the center O to AB meets AB exactly at P , which implies P must be the midpoint of AB .

- (b) **Uniqueness:** Yes, the chord is unique if P is strictly inside the circle and $P \neq O$. *Geometric Reasoning:* A chord is uniquely determined by its perpendicular distance and orientation from the center. For P to be the midpoint, the chord must be strictly perpendicular to the radial line segment OP . Through a given point P on a line OP , there exists exactly one perpendicular line in Euclidean geometry. Thus, there is exactly **1** such chord. (Note: If P were the center O , there would be infinitely many such chords, as every diameter is bisected by the center).

Solution to Question 3:

- (a) By the Fundamental Theorem of Calculus, $f(x) = \int_0^x f'(t)dt$. Using the Cauchy-Schwarz inequality on $[0, x]$: $|f(x)|^2 = \left(\int_0^x 1 \cdot f'(t)dt\right)^2 \leq \left(\int_0^x 1^2 dt\right) \left(\int_0^x (f'(t))^2 dt\right) \leq x \cdot 1 = x$. Thus $|f(x)| \leq \sqrt{x}$ for all $x \in [0, 1]$. Since $\sin x$ is bounded, the integral $\int_0^1 f(x) \sin x dx$ is absolutely bounded, so M is finite.

For the identity, we use Integration by Parts. Let $u = f(x)$ and $dv = \sin x dx$. We can choose the antiderivative $v = C - \cos x$. To make the boundary term vanish at $x = 1$, we choose $v = \cos 1 - \cos x$.

$$\int_0^1 f(x) \sin x dx = [f(x)(\cos 1 - \cos x)]_0^1 - \int_0^1 (\cos 1 - \cos x) f'(x) dx$$

Since $f(0) = 0$, the boundary evaluates to $f(1)(\cos 1 - \cos 1) - 0 = 0$. Reversing the sign inside the integral yields:

$$\int_0^1 f(x) \sin x dx = \int_0^1 (\cos x - \cos 1) f'(x) dx$$

- (b) By Cauchy-Schwarz on the new integral:

$$\int_0^1 (\cos x - \cos 1) f'(x) dx \leq \sqrt{\int_0^1 (\cos x - \cos 1)^2 dx} \sqrt{\int_0^1 (f'(x))^2 dx}$$

Since $\int (f')^2 \leq 1$, the maximum M is exactly $\left(\int_0^1 (\cos x - \cos 1)^2 dx\right)^{1/2}$.

$$M^2 = \int_0^1 (\cos^2 x - 2 \cos 1 \cos x + \cos^2 1) dx = \left[\frac{x}{2} + \frac{\sin 2x}{4} - 2 \cos 1 \sin x + x \cos^2 1 \right]_0^1$$

$$\Rightarrow M^2 = \frac{1}{2} + \frac{\sin 2}{4} - 2 \cos 1 \sin 1 + \cos^2 1 = \frac{1}{2} - \frac{3 \sin 2}{4} + \cos^2 1$$

$$M = \sqrt{\frac{1}{2} - \frac{3 \sin 2}{4} + \cos^2 1}$$

Equality holds in Cauchy-Schwarz when $f'(x)$ is proportional to $(\cos x - \cos 1)$. Since $\int (f')^2 = 1$, the unique function is obtained by integrating $f'(x) = \frac{\cos x - \cos 1}{M}$:

$$f(x) = \frac{\sin x - x \cos 1}{M}$$

- (c) We rewrite the expression $\int_0^1 f(x) dx - \lambda f(1)$ using Integration by Parts on the integral. Let $u = f(x)$ and $dv = 1 dx$. Choose $v = x + C$.

$$\int_0^1 f(x) dx = [f(x)(x + C)]_0^1 - \int_0^1 (x + C) f'(x) dx = f(1)(1 + C) - \int_0^1 (x + C) f'(x) dx$$

To absorb the $-\lambda f(1)$ term, we need $1 + C = \lambda \Rightarrow C = \lambda - 1$. Thus:

$$\int_0^1 f(x) dx - \lambda f(1) = - \int_0^1 (x + \lambda - 1) f'(x) dx$$

Applying Cauchy-Schwarz as before, the maximum is bounded by $\sqrt{\int_0^1 (x + \lambda - 1)^2 dx}$.

$$\int_0^1 (x + \lambda - 1)^2 dx = \left[\frac{1}{3} (x + \lambda - 1)^3 \right]_0^1 = \frac{1}{3} (\lambda^3 - (\lambda - 1)^3) = \lambda^2 - \lambda + \frac{1}{3}$$

The maximum value is explicitly:

$$\sqrt{\lambda^2 - \lambda + \frac{1}{3}}$$

Solution to Question 4:

- (a) Any polynomial $P(A)$ can be reduced to a degree of at most 2 by repeatedly applying the relation $A^3 = A$ (which implies $A^4 = A^2$, $A^5 = A$, etc.). Thus $P(A) = aI + bA + cA^2 = T(a, b, c)$. Because I, A, A^2 are explicitly given as linearly independent, this representation is unique.

Idempotent ($T^2 = T$): $T^2 = (aI + bA + cA^2)^2 = a^2I + 2abA + (b^2 + 2ac)A^2 + 2bcA^3 + c^2A^4$. Using $A^3 = A$ and $A^4 = A^2$: $T^2 = a^2I + (2ab + 2bc)A + (b^2 + 2ac + c^2)A^2$. Equating coefficients with $aI + bA + cA^2$ gives the system: 1) $a^2 = a \implies a = 0$ or 1 . 2) $2b(a + c) = b$. 3) $b^2 + 2ac + c^2 = c$. If $a = 0$: $2bc = b \implies b = 0$ or $c = 1/2$. If $b = 0 \implies c^2 = c \implies c \in \{0, 1\}$. If $c = 1/2 \implies b^2 = 1/4 \implies b = \pm 1/2$. If $a = 1$: $2b(1 + c) = b \implies b = 0$ or $c = -1/2$. If $b = 0 \implies c^2 + c = 0 \implies c \in \{0, -1\}$. If $c = -1/2 \implies b^2 = 1/4 \implies b = \pm 1/2$. Idempotent triples: $(0, 0, 0), (0, 0, 1), (0, \pm \frac{1}{2}, \frac{1}{2}), (1, 0, 0), (1, 0, -1), (1, \pm \frac{1}{2}, -\frac{1}{2})$.

Involutory ($T^2 = I$): Equating coefficients to $1I + 0A + 0A^2$: 1) $a^2 = 1 \implies a = \pm 1$. 2) $2b(a + c) = 0$. 3) $b^2 + 2ac + c^2 = 0$. If $a = 1$: $b(1 + c) = 0 \implies b = 0$ or $c = -1$. If $b = 0 \implies c^2 + 2c = 0 \implies c \in \{0, -2\}$. If $c = -1 \implies b^2 - 2 + 1 = 0 \implies b = \pm 1$. If $a = -1$: $b(-1 + c) = 0 \implies b = 0$ or $c = 1$. If $b = 0 \implies c^2 - 2c = 0 \implies c \in \{0, 2\}$. If $c = 1 \implies b^2 - 2 + 1 = 0 \implies b = \pm 1$. Involutory triples: $(1, 0, 0), (1, 0, -2), (1, \pm 1, -1), (-1, 0, 0), (-1, 0, 2), (-1, \pm 1, 1)$.

- (b) **Nilpotent**: We want $T^k = 0$. We can test T against factors of $A^3 - A = 0$. Multiply $T = aI + bA + cA^2$ by $(A^2 - I)$: $T(A^2 - I) = aA^2 - aI + bA^3 - bA + cA^4 - cA^2$. Using $A^3 = A, A^4 = A^2$, this simplifies to $a(A^2 - I)$. Thus, by induction, $T^k(A^2 - I) = a^k(A^2 - I)$. If $T^k = 0$, then $a^k(A^2 - I) = 0 \implies a = 0$ (since $A^2 - I \neq 0$). Now set $a = 0$, so $T = bA + cA^2$. Multiply T by $(A^2 - A)$: $T(A^2 - A) = bA^3 - bA^2 + cA^4 - cA^3 = bA - bA^2 + cA^2 - cA = (c - b)(A^2 - A)$. Thus $T^k(A^2 - A) = (c - b)^k(A^2 - A) \implies c - b = 0 \implies c = b$. Multiply T by $(A^2 + A)$: $T(A^2 + A) = bA^3 + bA^2 + cA^4 + cA^3 = bA + bA^2 + cA^2 + cA = (b + c)(A^2 + A)$. Thus $T^k(A^2 + A) = (b + c)^k(A^2 + A) \implies b + c = 0 \implies c = -b$. The only way $c = b$ and $c = -b$ is if $b = c = 0$. Thus the only nilpotent triple is $(0, 0, 0)$.

Inverse: We want $T' = a'I + b'A + c'A^2$ such that $TT' = I$. Expanding TT' and applying $A^3 = A, A^4 = A^2$: $TT' = aa'I + (ab' + a'b + bc' + b'c)A + (ac' + a'c + bb' + cc')A^2 = I$. Equating coefficients (since I, A, A^2 are linearly independent): 1) $aa' = 1 \implies a' = 1/a$. (Note: $a \neq 0$ because if $a = 0$, $T(A^2 - I) = 0$, meaning T is not invertible). 2) $(a + c)b' + bc' = -b/a$. 3) $bb' + (a + c)c' = -c/a$. Multiply (2) by $(a + c)$ and (3) by b , then subtract to eliminate c' :

$((a+c)^2 - b^2)b' = -b(a+c)/a + bc/a = -b \implies b' = \frac{-b}{(a+c)^2 - b^2}$. Substitute b' back into (3) to solve for c' : $c' = \frac{-c/a - bb'}{a+c} = \frac{-c/a + \frac{b^2}{(a+c)^2 - b^2}}{a+c} = \frac{-c((a+c)^2 - b^2) + ab^2}{a(a+c)((a+c)^2 - b^2)}$. Simplifying the numerator gives $-ca^2 - 2ac^2 - c^3 + cb^2 + ab^2 = -c(a+c)^2 + (a+c)b^2$. Factoring out $(a+c)$ and dividing yields $c' = \frac{a+c}{(a+c)^2 - b^2} - \frac{1}{a}$. Formulas: $\mathbf{a}' = \frac{1}{\mathbf{a}}$, $\mathbf{b}' = \frac{-\mathbf{b}}{(\mathbf{a}+\mathbf{c})^2 - \mathbf{b}^2}$, $\mathbf{c}' = \frac{\mathbf{a}+\mathbf{c}}{(\mathbf{a}+\mathbf{c})^2 - \mathbf{b}^2} - \frac{1}{\mathbf{a}}$.

- (c) We need a polynomial $q(x) = c_0 + c_1x + c_2x^2$ such that $q(A) = A^2$. $c_0I + c_1A + c_2A^2 = A^2$. By the strict linear independence of I, A, A^2 , the coefficients must match exactly: $c_0 = 0, c_1 = 0, c_2 = 1$. Thus, $q(x) = x^2$ is the unique polynomial of degree ≤ 2 . We verify $q(q(A)) = q(A)$ using $A^3 = A$: $q(q(A)) = q(A^2) = (A^2)^2 = A^4 = A \cdot A^3 = A \cdot A = A^2$. Since $q(A) = A^2$, we have $q(q(A)) = q(A)$. Unique polynomial: $\mathbf{q}(\mathbf{x}) = \mathbf{x}^2$.

Solution to Question 5:

- (a) Let $d = \gcd(a+b, a^2 - ab + b^2)$. We can rewrite $a^2 - ab + b^2 = (a+b)^2 - 3ab$. Since $d \mid (a+b)$, d must divide $(a+b)^2$. Consequently, d must divide $3ab$. Since $\gcd(a, b) = 1$, $a+b$ shares no prime factors with a or b . Therefore, d shares no prime factors with a or b , meaning $\gcd(d, ab) = 1$. This forces $d \mid 3$, so $d \in \{1, 3\}$.

For $Q(a, b) = a+b$: $2a^2 - 2ab + 2b^2 = 2a + 2b \implies (a-b)^2 + (a-1)^2 + (b-1)^2 = 2$. The sum of three squares equals 2 implies the squares are 1, 1, 0. If $(a-b)^2 = 0 \implies a = b \implies a = 2, b = 2$. Wait, $\gcd(2, 2) \neq 1$, but the equation asks for *all* positive integer solutions, regardless of coprime status. If $(a-1)^2 = 0 \implies a = 1 \implies (b-1)^2 = 1 \implies b = 2$. If $(b-1)^2 = 0 \implies b = 1 \implies (a-1)^2 = 1 \implies a = 2$. Solutions for $Q(a, b) = a+b$: $(1, 2), (2, 1), (2, 2)$.

For $Q(a, b) = 3(a+b)$: $2a^2 - 2ab + 2b^2 = 6a + 6b \implies (a-b)^2 + (a-3)^2 + (b-3)^2 = 18$. Possible sums of squares ≤ 18 yielding 18 are $9+9+0$ and $16+1+1$. Testing $16+1+1$ yields no integer solutions because the parity of differences contradicts $(a-b)^2 = 16$. Using $9+9+0$: If $(a-b)^2 = 0 \implies a = b = 6$. If $(a-3)^2 = 0 \implies a = 3, b = 6$. If $(b-3)^2 = 0 \implies b = 3, a = 6$. Solutions for $Q(a, b) = 3(a+b)$: $(3, 6), (6, 3), (6, 6)$.

- (b) We are given $\gcd(a, b) = 1$ and $Q(a, b)^2 \mid (a^3 + b^3)$. We know $a^3 + b^3 = (a+b)Q(a, b)$. Thus, $Q(a, b)^2 \mid (a+b)Q(a, b) \implies Q(a, b) \mid (a+b)$. This means the greatest common divisor of $Q(a, b)$ and $a+b$ is $Q(a, b)$ itself. But from part (a), because a, b are coprime, their GCD can only be 1 or 3. Thus, $Q(a, b) \in \{1, 3\}$.

If $Q(a, b) = 1$: $a^2 - ab + b^2 = 1$. For positive integers, $a = 1, b = 1$. If $Q(a, b) = 3$: $a^2 - ab + b^2 = 3$. If $a = 1 \implies b^2 - b - 2 = 0 \implies b = 2$. If $a = 2 \implies b^2 - 2b + 1 = 0 \implies b = 1$. Solutions: $(1, 1), (1, 2), (2, 1)$. All are coprime and satisfy the condition.

- (c) We rewrite the equation $a^2 - ab + b^2 = n$ by completing the square:

$$\frac{3}{4}a^2 + \left(b - \frac{a}{2}\right)^2 = n$$

Since the squared term $(b - a/2)^2$ is always non-negative, it must be true that:

$$\frac{3}{4}a^2 \leq n \implies a^2 \leq \frac{4n}{3} \implies a \leq \sqrt{\frac{4n}{3}}$$

By symmetry in the original equation, completing the square for b yields the same logic, meaning $b \leq \sqrt{4n/3}$. Since both a and b are bounded strictly by a finite number dependent only on n , the total number of positive integer coordinate pairs (a, b) within this finite square is finite.

Solution to Question 6

- (a) Since f is continuous on $[0, 1]$ and differentiable on $(0, 1)$, we apply the Lagrange Mean Value Theorem on the interval $[0, 1]$. There exists some $c \in (0, 1)$ such that:

$$f'(c) = \frac{f(1) - f(0)}{1 - 0} = \frac{1 - 0}{1} = 1$$

- (b) We want to find the largest K . We use Taylor's Theorem with remainder. Expand $f(1/2)$ from $x = 0$:

$$f(1/2) = f(0) + \frac{1}{2}f'(0) + \frac{(1/2)^2}{2}f''(c_1) = \frac{1}{8}f''(c_1) \quad \text{for some } c_1 \in (0, 1/2)$$

Expand $f(1/2)$ from $x = 1$:

$$f(1/2) = f(1) - \frac{1}{2}f'(1) + \frac{(-1/2)^2}{2}f''(c_2) = 1 + \frac{1}{8}f''(c_2) \quad \text{for some } c_2 \in (1/2, 1)$$

Equating the two expressions for $f(1/2)$:

$$\frac{1}{8}f''(c_1) = 1 + \frac{1}{8}f''(c_2) \implies f''(c_1) - f''(c_2) = 8$$

By the triangle inequality, $|f''(c_1)| + |f''(c_2)| \geq |f''(c_1) - f''(c_2)| = 8$. If the sum of two absolute values is ≥ 8 , the maximum of the two must be at least 4. Thus, there exists some $c \in \{c_1, c_2\}$ strictly inside $(0, 1)$ where $|f''(c)| \geq 4$. The strictly largest bound guaranteed is $\boxed{K = 4}$.

- (c) To prove the bound $K = 4$ is sharp, we must explicitly construct a valid, strictly twice-differentiable function $g(x)$ where the absolute second derivative never exceeds $4 + \epsilon$.

The "ideal" shape for the derivative is a triangle wave $h'(x) = 2 - 2|2x - 1|$, which gives $|h''(x)| = 4$ but fails to be differentiable at $x = 1/2$. We can smooth the sharp corner of the absolute value using the hyperbola $\sqrt{(2x - 1)^2 + \delta^2}$, where $\delta > 0$ is a small smoothing parameter. Define our smooth derivative as:

$$g'(x) = c_\delta \left(\sqrt{1 + \delta^2} - \sqrt{(2x - 1)^2 + \delta^2} \right)$$

Note that $g'(x)$ is smooth everywhere on $(0, 1)$, and $g'(0) = g'(1) = 0$, satisfying the boundary conditions for the derivative. To ensure $g(1) - g(0) = 1$, we must scale the function so its integral is exactly 1. Let $I(\delta)$ be the unscaled integral:

$$I(\delta) = \int_0^1 \left(\sqrt{1 + \delta^2} - \sqrt{(2x - 1)^2 + \delta^2} \right) dx$$

We set the scaling constant $c_\delta = \frac{1}{I(\delta)}$. As $\delta \rightarrow 0$, the function $\sqrt{(2x-1)^2 + \delta^2}$ converges uniformly to $|2x-1|$. Thus, the integral evaluates to:

$$\lim_{\delta \rightarrow 0} I(\delta) = \int_0^1 (1 - |2x-1|) dx = \frac{1}{2}$$

This implies that $\lim_{\delta \rightarrow 0} c_\delta = 2$.

Now, let us bound the second derivative $g''(x)$:

$$g''(x) = \frac{d}{dx} \left[c_\delta \left(\sqrt{1+\delta^2} - \sqrt{(2x-1)^2 + \delta^2} \right) \right] = -c_\delta \frac{2(2x-1)}{\sqrt{(2x-1)^2 + \delta^2}}$$

We observe that for all real y , the fraction $\frac{|y|}{\sqrt{y^2 + \delta^2}} < 1$. Letting $y = 2x-1$, we get a strict global bound:

$$|g''(x)| = 2c_\delta \frac{|2x-1|}{\sqrt{(2x-1)^2 + \delta^2}} < 2c_\delta$$

Since $\lim_{\delta \rightarrow 0} 2c_\delta = 4$, for any arbitrarily small $\epsilon > 0$, we can choose $\delta > 0$ small enough such that $2c_\delta \leq 4 + \epsilon$. Consequently, for this explicitly chosen $g(x)$, we have $|g''(x)| < 4 + \epsilon$ for all $x \in (0, 1)$, establishing that the bound $K = 4$ cannot be improved.